

PROFESSIONAL SUMMARY

PhD Candidate in Computer Engineering and innovative educator with extensive experience instructing and mentoring large, diverse undergraduate and graduate cohorts. Proficient in C++ and Python, with demonstrated excellence in translating complex technical concepts into accessible, experiential learning opportunities. Committed to inclusive pedagogy, curriculum development, and bridging human-centered inquiry with data ethics to ensure student access and success. Proven track record in designing curriculum for software engineering, deep learning, and foundational computer science courses.

EDUCATION

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| 08/2026 | DOCTOR OF PHILOSOPHY, COMPUTER ENGINEERING (<i>Expected</i>)
Iowa State University (ISU), Ames, IA
Major Professors: Goce Trajcevski, Ashfaq Khokhar
Thesis: Constraint-Aware Scalable Discovery of Persistent Spatio-Temporal Patterns |
| 12/2024 | MASTER OF SCIENCE, COMPUTER ENGINEERING
Iowa State University (ISU), Ames, IA
Major Professors: Goce Trajcevski
Specialization: Software Systems |
| 06/2020 | BACHELOR OF SCIENCE, COMPUTER SCIENCE
Lahore University of Management Sciences (LUMS), Lahore, Pakistan
Major Professor: Muhammad Fareed Zaffar
BS Thesis: Authorship Attribution and Obfuscation |

TEACHING EXPERIENCE

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| Spring 2026 | Guest Lecturer & Graduate Teaching Assistant – ISU |
| Spring 2025 | Software Evolution and Maintenance <ul style="list-style-type: none">• Facilitated the study of long-term software lifecycles, emphasizing systematic defect analysis and debugging.• Mentored undergraduate senior students in tracing and understanding large, complex codebases.• Evaluated professional-grade written reports and oral presentations, providing feedback on technical communication and experimental studies of software maintenance. |

Fall 2025

Lab Instructor & Graduate Teaching Assistant – ISU

Introduction to Computer Engineering and Problem Solving

- Delivered foundational instruction to 150+ first-year students, introducing core computer engineering principles through project-based learning like algorithmic problem solving, C/C++ programming, and technical report writing.
- Guided small and large student groups through the full lifecycle of computer-based projects.
- Focused on developing interactive skills and professional presentation standards through human-centered inquiry, helping students transition from high school to rigorous engineering methodologies.
- Ensured timely and detailed feedback to students, helping them understand their progress well.

Spring 2025

Spring 2024

Lab Instructor & Graduate Teaching Assistant – ISU

Software Tools for Large-Scale Data Analysis

- Led weekly laboratory sessions, by introducing content with a mini lecture each time, for 30+ graduate and/or undergraduate students, on external memory processing, stream processing, large-scale parallelism (Hadoop/Spark/Pig Latin), and data interchange formats.
- Employed a project-based learning framework to bridge the gap between theoretical data science and practical software engineering, satisfying experiential education goals while teaching high-volume data management.
- Ensured timely delivery of all grades and feedback for all students to help them understand their errors.

Fall 2024

Lab Instructor & Graduate Teaching Assistant – ISU

Digital Logic Circuits (Fall 2024)

- Supervised laboratory sessions, by introducing content with a mini lecture each time, for 25+ undergraduate sophomore/junior students, on Boolean algebra, Finite State Machines (FSMs), Verilog/VHDL hardware description languages, and programmable logic devices (FPGA).
- Instructed students on computer-aided schematic capture and simulation tools.
- Provided hands-on guidance for building arithmetic circuits and sequential logic, ensuring students could translate abstract logic minimization into functional physical hardware.

Fall 2023

Graduate Teaching Assistant – ISU

Software Analysis & Verification for Safety & Security (Fall 2023)

- Guided 80+ graduate and/or undergraduate senior students through the mathematical foundations of software safety, threat modeling, and vulnerability categorization for cyber-physical systems by holding regular office hours and tutorials.
- Facilitated case studies on large-scale systems software, helping students implement verification algorithms and navigate the evolving landscape of security mitigation techniques.
- Provided technical mentorship on program comprehension and the use of professional analysis tools.
- Graded all course content for the students, ensuring timely delivery of feedback and grades.

Fall 2021

Spring 2021

Guest Lecturer & Lab Instructor – ISU

Introduction to Spreadsheets and Databases (Fall 2021, Spring 2021)

- Managed instructional logistics for a high-enrollment foundational course of 200+ undergraduate students, specifically focusing on building critical digital literacies for non-technical learners.
- Delivered lab lectures on Microsoft Excel and Access, translating data management concepts into accessible, experiential tutorials.
- Supported a diverse cohort by developing automated grading scripts for rapid feedback, prioritizing an inclusive, accessible learning environment to ensure no student faced technical barriers.

Spring 2020

Guest Lecturer and Undergraduate Teaching Assistant – LUMS

Deep Learning

- Delivered guest lectures and led project-based laboratory sessions on implementing neural network architectures for graduate and undergraduate cohorts.
- Simplified complex mathematical concepts into accessible modules, covering backpropagation, stochastic gradient descent, and loss function optimization.
- Conducted specialized tutorials on hyperparameter tuning and model evaluation metrics, guiding students through the practical, applied AI challenges required for success in their final projects.

Fall 2019

Undergraduate Teaching Assistant – LUMS

Discrete Mathematics

- Held tutorials for undergraduate students on the mathematical foundations of computer science, including Set Theory, Logic, Combinatorics, and Graph Theory.
- Held dedicated recitation sessions to help students master the art of formal proofs and induction.

- Developed practice problem sets and exam materials that emphasized the application of discrete math in algorithm design and data structures.

Spring 2019
Fall 2018

Lead Undergraduate Lab Instructor – LUMS

Computational Problem Solving

- Managed instructional logistics for a high-enrollment foundational course.
- Led weekly recitations and extra tutorials designed to help a diverse student body master algorithmic thinking and syntax.
- Assisted in the creation of interactive class activities that broke down complex engineering problems into manageable logical steps.
- Focused on teaching students how to move from a conceptual flowchart to a functional implementation in C++.
- Graded exams and programming assignments for 200+ students, providing detailed feedback to bridge the gap between theoretical logic and practical debugging.
- Implemented "early alert" tutorials for students struggling with core programming constructs, significantly improving overall class performance.

MENTORING EXPERIENCE

01/2022 – 05/2023 **PROJECT MENTORING – ISU (SPRING/FALL 2022, SPRING 2023)**

- Mentored multiple senior-year capstone teams (100+ students) through the full Software Development Life Cycle.
- Guided students in making critical architectural decisions, selecting appropriate technology stacks, and implementing Agile project management methodologies.
- Facilitated weekly "sprint" reviews to resolve technical roadblocks, conduct code reviews, and ensure students adhered to industry standards for software quality, security, and documentation.
- Successfully mentored a student group in their senior-year project that led to a co-authored demo paper at **ACM SIGSPATIAL 2024**, showcasing my ability to elevate undergraduate coursework to the level of internationally recognized research.

INDUSTRY EXPERIENCE

- 05/2025 – 08/2025 **ENGINEERING DATA SCIENCE INTERN**, Altair Engineering, Troy, MI
- Built and deployed end-to-end PyTorch and TensorFlow pipelines supporting training, evaluation, and GPU-accelerated inference within a production simulation platform.
 - Developed and optimized scalable architectures for PhysicsAI models, achieving a 10x reduction in Mean Absolute Error (MAE) for complex spatio-temporal physical simulation data.
 - Engineered real-world case studies from these industry pipelines, directly applying this experience to teach scalable modeling to senior engineering students.
- 05/2018 – 06/2018 **SOFTWARE DEVELOPER**, Taar Consulting, Lahore, Pakistan
- Designed and seamlessly integrated a Customer Feedback Plugin for Retail Pro Prism, capturing feedback from over 1,000 daily transactions to improve business operations.

RESEARCH EXPERIENCE

- 06/2022 – Present **GRADUATE RESEARCH ASSISTANT – ISU**
- SDC Framework (Accepted - IEEE MDM 2026)**
- Designed the Semantic Diversity in Convoys (SDC) framework to advance diversity-aware clustering in complex dynamic environments (e.g., molecular dynamics, biological movement).
 - Implemented Brute-Force, ED-SDC, and Grid-SDC algorithms, leveraging indexing and spatial partitioning to achieve 17x to 65x computational speedups.
 - Integrated transformers and embedding-based similarity measures to efficiently process and identify persistent spatio-temporal patterns.
- Motif Mining and Scientific ML (Pending Submission - 2026)**
- Developing large-scale motif-mining pipelines leveraging geometric descriptors, spatio-temporal statistics, and learned embeddings.
 - Identify aromatic ring-pair configurations predictive of HB persistence - enabling new scientific ML approaches for event prediction and simulation understanding.

Bond-Aware Relaxed Moving Clusters (BARMC) (SIGSPATIAL 2024, TSAS 2026)

- Introduced, developed, and implemented the Bond-Aware Relaxed Moving Clusters (BARMC) algorithm.
- Optimized the detection of key interactions within complex datasets by explicitly integrating spatial proximity metrics with domain-specific semantic rules.
- Enhanced traditional clustering techniques, successfully achieving a ~1.7x reduction in computation time for complex trajectory data.

Automatic Detection of Hydrogen Bonds (HBs) (SIGSPATIAL 2024)

- Directed and mentored an undergraduate research team through the full lifecycle of an applied software engineering project, focusing on the real-time detection and analysis of dynamic formations in simulations.
- Facilitated experiential learning by guiding students to automate the identification of long-lasting interactions, successfully bridging theoretical computer science concepts with practical reductions in computational overhead.
- Supervised the students' development of robust visualization components for data analysis, cultivating their technical problem-solving skills and culminating in a co-authored publication at SIGSPATIAL 2024.

Compressing Molecular Dynamics (MD) Data (ADBIS 2022)

- Developed domain-specific trajectory compression algorithm for MD simulations, achieving an 80% reduction in storage requirements without sacrificing query effectiveness.
- Implemented the algorithm to optimize within-distance query processing, achieving a 3x speedup in data retrieval for complex simulations.
- Led to the publication of an award-winning paper in Advances in Databases and Information Systems (ADBIS 2022).

01/2021 – 05/2022 **INDEPENDENT RESEARCHER – ISU**

Large-Scale Data Analysis:

- Leveraged Java and Hadoop MapReduce API for large-scale data analysis, focusing on text analytics.
- Developed and optimized an algorithm to efficiently identify the top 10 most frequent bigrams in extensive text corpora, achieving a 2x speedup in data processing.
- Also engineered a Word Count algorithm on Shakespeare and Gutenberg datasets, confirming the program's scalability and robustness.

Sarcasm Detection in News Headlines:

- Leveraged Python and advanced machine learning algorithms, including LSTM, SVM and Decision Trees, to develop a sarcasm detection model for news headlines sourced from a Kaggle dataset.
- Achieved a high classification precision of 84% and recall of 82%, employing dense vector representations for tokenized and padded input, improving the computer's ability to identify sarcasm in textual data.

Deep Convolutional GAN for High-Quality Emoji Generation

- Successfully implemented and trained a Deep Convolutional Generative Adversarial Network (DCGAN) for emoji generation, achieving optimal performance metrics with a Discriminator Loss of 0.426 and a Generator Loss of 1.665 after 2500 epochs.
- Applied research-backed stabilization techniques to refine the model.
- This applied AI project serves as an ideal framework for master's level curriculum development and student-led experiential learning.

Classification using Layer-Optimized Convolutional Neural Networks

- Executed a two-phase project to classify the CIFAR-10 dataset using Convolutional Neural Networks (CNNs).
- The first phase included the manual implementation of a CNN, overcoming the inherent challenges of image classification.
- The second phase utilized Keras to experiment with advanced layers including Pooling, Dropout, and Batch Normalization.
- Conducted a model comparison, achieving a high classification accuracy of 72% with an optimized, multi-layered CNN model.

PUBLICATIONS

PEER-REVIEWED JOURNAL PUBLICATIONS

1. **Shamail, A.**, Anowar, M., Trajcevski, G., Khokhar, A., Murad, S., Jameson, C., 2026. Bond-Aware Moving Clusters of Atomic Trajectories with Relaxed Persistency. *ACM Trans. Spatial Algorithms Syst.* 12, 2, Article 8 (June 2026), 27 pages.
<https://doi.org/10.1145/3804446>
2. Anowar, M., **Shamail, A.**, Wang, X., Trajcevski, G., Murad, S., Jameson, C. J., & Khokhar, A. (2024). Compressing generalized trajectories of molecular motion for efficient detection of chemical interactions. *Information Systems*, 125, 102426.
<https://doi.org/10.1016/j.is.2024.102426>
3. Mubin, O., Alhajjar, F., **Shamail, A.**, Shahid, S., Simoff, S. The new norm: Computer Science conferences respond to COVID-19. *Scientometrics* 126, 1813–1827 (2021).
<https://doi.org/10.1007/s11192-020-03788-9>

CONFERENCE PROCEEDINGS

1. **Shamail, A.**, Trajcevski, G., Khokhar, A., Zufle, A. 2026. Semantically Diverse Convoys, 27th IEEE International Conference on Mobile Data Management, June 29th - July 2nd, 2026. Athens, Greece

2. **Shamail, A.**, Anowar, M., Trajcevski, G., Khokhar, A., Murad, S., Jameson, C., Bond-Aware Moving Clusters of Atomic Trajectories with Relaxed Persistency. In Proceedings of the 32nd ACM International Conference on Advances in Geographic Information Systems (SIGSPATIAL '24). Association for Computing Machinery, New York, NY, USA, 605–608. <https://doi.org/10.1145/3678717.3691298>
3. Anowar, M., **Shamail, A.**, Albertsen, A., Hall, B., Thielen, T., Riemersma, B., Trajcevski, G. Detecting and Visualizing Bond-Forming Convoys in Atomic and Molecular Trajectories. In Proceedings of the 32nd ACM International Conference on Advances in Geographic Information Systems (SIGSPATIAL '24). Association for Computing Machinery, New York, NY, USA, 657–660. <https://doi.org/10.1145/3678717.3691264>
4. Anowar, M., **Shamail, A.**, Wang, X., Trajcevski, G., Murad, S., Jameson, C. J., & Khokhar, A. (2022). Generalization Aware Compression of Molecular Trajectories. In: Chiusano, S., Cerquitelli, T., Wrembel, R. (eds) Advances in Databases and Information Systems. ADBIS 2022. Lecture Notes in Computer Science, vol 13389. Springer, Cham. https://doi.org/10.1007/978-3-031-15740-0_20

SERVICE AND LEADERSHIP

SERVICE TO DISCIPLINE

11/2024

Volunteer, ACM SIGSPATIAL 2024 – Atlanta, Georgia

- Contributed to the successful execution of a top-tier international conference by managing the registration desk, coordinating logistics for 300+ attendees, and facilitating technical sessions.
- Helped with the setup of research poster sessions, ensuring optimal flow and visibility for presenters.
- Acted as a point of contact for attendees, providing venue guidance and technical assistance during plenary and break-out sessions.

SERVICE TO CAMPUS

05/2023 – 05/2024

President - Pakistan Student Association (PSA) – ISU

- Led community engagement initiatives, notably organizing a Pakistan Independence Day event that attracted over 80 attendees.
- Oversaw budget management and financial planning, utilizing data-driven approaches to optimize resource allocation.
- Designed and executed a social media campaign, leveraging real-time data analytics to refine strategies.
- This ultimately helped raise \$20,000 for flood relief efforts, which supported 15 million affected individuals in Pakistan.
- Participated in, and cooked for, the International Culture and Food Fest attended by more than 300 attendees at ISU.
- Volunteered weekly at ISU Dining services to help them cater to the dining demands of the students.

05/2019 – 05/2020 **Vice President - Index: Design and Innovation Society – LUMS**

- Initiated the first-ever Design Innovation Event, using quantitative metrics to identify key areas of interest and measure outcomes.
- Mentored a cohort of 50 students, employing performance indicators and KPIs to track progress and assess impact.
- Successfully organized UX Pakistan, an international design conference that attracted approximately 500 global attendees annually from 2018 to 2020. Additionally, hosted the National Design Awards in Pakistan in 2018.

HONORS AND AWARDS

1. Recipient of the **NSF (National Science Foundation) Student Travel Grant**, awarded through a competitive selection process to support conference attendance and professional development.
2. Best Paper award for the research: Generalization Aware Compression of Molecular Trajectories at **ADBIS 2022**
3. Awarded the title of **“Preparing Future Faculty Associate”** by Iowa State University in 2025

SPOKEN LANGUAGES

English (fluent), Urdu (native), Punjabi (intermediate)

SOFTWARE AND CODING

Data Analysis and Machine Learning: TensorFlow, PyTorch, Sklearn, Keras, Pandas, Numpy, Apache Spark, Apache Flink, Hadoop, Pig Latin, SQL, MATLAB, SQL Server, MDX, MySQL.

Core Languages: Python, C++, Objective C, JavaScript/ReactJS.

Familiarity: GoLang, Haskell, Java, PHP.

Miscellaneous: LaTeX, GitHub, Trajectory Analysis, Slack, Discord, MS Excel, MS Access.

REFERENCES

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